FutureSchool High School Management System — Complete Architecture & Flow

This document details the complete end-to-end architecture, role-based interaction models, database schema, and operational data flows of the **FutureSchool High School Management System**.

1. High-Level System Architecture

The application implements a modern **Three-Tier Architecture**:

2. Actor & Role-Based Access Control (RBAC)

The system enforces strict multi-tenancy access control partitioned across 5 distinct school roles:

3. End-to-End Authentication & Authorization Lifecycle

4. Key Business Flows

Flow A: Student Admission & Automatic Credential Provisioning

Flow B: Parent Multi-Child / Sibling Switcher

Flow C: Complete Academic & Examination Lifecycle

Flow D: Fee Billing & Cashier Collection Lifecycle

5. Relational Database Schema (PostgreSQL 17)

The backend schema comprises **24 relational tables** organized into core domain modules:

6. Technical Stack & Deployment Configuration

Component	Technology	Version	Port / Path	Purpose
Frontend UI	React, TypeScript, Vite, Tailwind CSS	18.3.1	`http://localhost:3000`	Single-page UI application
Backend API	Spring Boot, Spring Data JPA, Spring Security	3.3.4	`http://localhost:5000`	REST API service layer
Database	PostgreSQL	17.0	`localhost:5432/futureschool`	Relational persistence
Authentication	JJWT (io.jsonwebtoken)	0.12.5	Stateless Bearer Auth	Token verification & RBAC
API Docs	SpringDoc OpenAPI (Swagger UI)	2.6.0	`http://localhost:5000/swagger-ui/index.html`	Interactive API explorer
API Client	Postman Collection (v2.1)	JSON	`D:\Student Management\FutureSchool_Postman_Collection.json`	Automated API testing